

SECOM Yield Optimization

Predicting low-yield wafers in semiconductor packaging & test · Industrial IoT

The question Which process signals predict low-yield (fail) wafers, and how can the fab use them to raise yield and cut per-unit cost?

~14 : 1

pass : fail imbalance

~41,951

missing sensor values

590

anonymized sensors

What we did

A clean, reproducible, leakage-safe pipeline: drop unreliable and near-constant sensors → median-impute → winsorize → standardise → handle imbalance with class weights and SMOTE → compare Logistic Regression and Random Forest, judged on recall and PR-AUC for the fail class (not accuracy). Feature importance is surfaced as a yield Pareto. The same aggregation is also expressed at fab scale in Pig, Hive (Spark SQL and real Hadoop+Hive) and DuckDB.

What we found

- **Accuracy is a trap** With ~14:1 imbalance a 'predict pass' model scores ~93% and catches zero failures — so we optimise recall / PR-AUC.
- **The signal is concentrated** A vital few sensor IDs carry most of the predictive power; the other ~570 are largely noise.
- **Failure is multivariate; gaps are structural** Top sensors overlap heavily (monitor a set, not one line); missingness marks sensors that were offline — a fixable data-quality issue.

Recommendations (highest-leverage first)

- 1 Put the vital-few sensor IDs under tightened SPC with automated alarms — the cheapest yield win.
- 2 Deploy the model as a recall-tuned early-warning screen; keep an engineer in the loop, don't auto-scrap.
- 3 Fix the offline-sensor instrumentation gaps at source.
- 4 Confirm top signals with designed experiments before changing recipes.
- 5 Retrain on a rolling cadence; use active learning to grow the scarce failure set.

Takeaway Catch the rare failure, watch the vital few — and the data pipeline becomes a durable lever on yield and per-unit cost.